



DIFFERENTIATION DOSSIER · M42 REVIEW · JUNE 2026

Aethelred x M42

Differentiation Dossier

How Aethelred differs from general-purpose Layer 1 networks, privacy chains, zkML projects, compute markets, and hyperscaler confidential computing - with every claim mapped to an artifact M42 inspects in its own pilot.

Prepared for	M42 technical and security reviewers
Prepared by	Ramesh Tamilselvan, Founder, Aethelred
Purpose	Evidence-anchored differentiation review
Delivery moment	After the Week 2 evidence room walkthrough

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This dossier compares Aethelred against each category an M42 reviewer will raise in an internal debate. The standard for inclusion is that every claim maps to a pilot artifact or public repository check M42 can verify directly.

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How to Read This Document

Every differentiation claim below is mapped to an artifact M42 inspects during its own pilot. The standard for inclusion is: if M42 cannot verify the claim from a pilot artifact or the public repository, the claim does not belong in this dossier. Where a capability is roadmap rather than current, it is labeled as such.

Delivery moment: hand to M42 technical and security reviewers after the Week 2 evidence room walkthrough, when they have just inspected the artifacts this document references.

The One-Paragraph Difference

General-purpose Layer 1s settle transactions and ask applications to bring their own trust. Aethelred settles evidence of verified AI computation: the unit of value on the chain is a Digital Seal binding a model hash, a workload policy (jurisdiction, data class, consent, retention), a hardware TEE attestation, input/output commitments, and a zkML proof where tractable - signed by validators whose work is itself useful computation. For a buyer like M42, the difference is not throughput or fees; it is that the thing M42's customers and regulators need to inspect is the chain's native object, not an application-layer convention bolted onto a generic ledger.

Category 1 - General-Purpose Layer 1s (Ethereum, Solana, Avalanche, generic Cosmos chains)

Dimension	General-purpose L1	Aethelred	Pilot artifact that shows it
Native object	Token transfer / contract state	Evidence bundle with Digital Seal	evidence-bundle-<job_id>.json - schema-validated, per accepted case
What validators do	Secure consensus; energy/stake spent on ordering	Proof of Useful Work: rewards for verified AI and scientific computation	Protocol design docs; validator fields in every bundle

Dimension	General-purpose L1	Aethelred	Pilot artifact that shows it
Confidential data	Public-by-default; confidential workloads need L2s or off-chain systems	Confidential-compute lanes with policy metadata fixed pre-execution	sandbox.json data-boundary controls; workload-pack.json policy block
Evidence rejection	Application's problem	Protocol-level: fallback, wrong-hash, replayed, or single-evidence submissions are rejected	negative-control-results.json - 6/6 rejections, run in M42's own pilot
Healthcare fit	Generic; HIPAA/UAE-data-class semantics live entirely in app code	Jurisdiction, consent scope, and retention are first-class policy fields	Policy metadata inside each evidence bundle
Post-quantum posture	Largely classical cryptography today	PQC-aligned signatures/commitments (NIST FIPS 203/204/205 alignment) for decades-long evidence validity	Crypto suite documented in evidence schema and SDK

The practical consequence: to give M42 what the pilot gives, a general-purpose L1 would need M42 (or a vendor) to build and operate the TEE orchestration, attestation verification, zkML pipeline, policy engine, evidence schema, and archive as custom application infrastructure - and the result would still be an application-layer convention a counterparty must be persuaded to trust, not a protocol guarantee.

Category 2 - Confidential-Compute Chains (privacy-focused L1s)

What they offer: TEE-backed private smart contract execution - confidentiality of contract state.

Where they stop short for M42's problem: confidentiality is necessary but not sufficient. M42's buyers need verifiability of AI execution - which model version ran, on what data class, in which jurisdiction, with what output commitment - not just that contract state was hidden. Privacy chains hide computation; Aethelred's purpose is to make computation provable while data stays protected. The evidence object, the model/circuit registries, the clinical workload policy semantics, and the negative-control rejection behavior are the product, not an application someone could write on a privacy chain.

Pilot artifacts: model registry (registry/measurements.json - active model hash binding), circuit registry (registry/circuits.json), and the wrong-model-hash negative control showing the chain rejecting a substituted model.

Category 3 - zkML and Verifiable-Inference Projects

What they offer: proof systems that a specific model produced a specific output - genuinely valuable, and Aethelred uses zkML as one evidence layer.

Where they stop short: zkML alone is a component, not an assurance system. Proof generation is tractable only for certain model sizes/architectures today; there is no data-sovereignty story, no jurisdiction policy, no archive/monitoring/operational layer, and no settlement venue a regulator can query years later. Aethelred's hybrid contract (TEE attestation plus zkML plus Digital Seal, with single-evidence fallback rejected for accepted paid work) is explicitly designed around zkML's current coverage limits, and says so per workload rather than implying universal proof.

Pilot artifacts: per-job proof-<job_id>.json and attestation-<job_id>.json sidecars; the documented proof-mode note in the business case (Section 7); the fallback-mode negative control rejection.

Category 4 - Decentralized Compute Markets

What they offer: cheaper raw GPU/CPU capacity through open markets - part of where Aethelred's cost advantage comes from on the supply side.

Where they stop short: they sell capacity, not assurance. No attestation contract, no policy lanes, no evidence settlement, no healthcare data-class semantics. M42 could rent cheap compute there and still have nothing to show a foreign regulator. Aethelred sits above competitively supplied capacity and adds the verification and evidence layer that makes regulated workloads sellable.

Pilot artifact: baseline vs verified economics in sandbox-run-summary.json - the measured overhead M42 pays for evidence, on top of competitively sourced compute.

Category 5 - Hyperscaler Confidential Computing (the real incumbent)

This is the comparison that matters most in an M42 internal debate, because Azure/GCP/AWS confidential compute is mature and M42 already uses hyperscalers.

Question an M42 buyer or regulator asks	Hyperscaler confidential compute	Aethelred
Can a foreign government verify the run without trusting the seller or its cloud vendor?	Attestation chains terminate in the vendor's and M42's own claims	Evidence settles on an independent ledger; verification requires trusting neither M42 nor a single vendor
Is the evidence durable and queryable in 10 years?	Logs and attestations live in vendor accounts under vendor retention	Digital Seals are durable chain objects with archive export (enterprise_export query paths)
Is pricing inspectable?	Internal allocation, negotiated discounts, opaque	Settlement on a transparent market; cost basis open to inspection
Sovereignty narrative	US hyperscaler trust root	UAE-domiciled (ADGM-planned) layer; in-region lanes
Sustainability narrative	Datacenter PPA claims	Proof of Useful Work: security spend is productive computation

Honest boundary: hyperscaler confidential compute is excellent infrastructure, and Aethelred TEE lanes can run on such hardware. The differentiation is the independent settlement and evidence layer above it - which is exactly the part the hyperscaler cannot supply, because it would be attesting to itself.

Category 6 - Building It In-House at M42

Covered in the business case (page 4) and repeated here because it recurs in every internal debate: a team of M42's scale could build a verification system, but it would be M42's system, and a counterparty would still be asked to trust M42's attestation of its own work. Independence is the product; an in-house system cannot supply it by definition. The same logic is why the investment structure (see corp-dev memo) caps any single party's consensus influence - including M42's.

Claims We Deliberately Do Not Make

Maintaining these boundaries is part of the differentiation - M42's reviewers will test them.

- No clinical-correctness claim: circuits bind IO, policy, and evidence commitments; they do not prove a free-text answer is clinically right. The clinical evaluation protocol measures that separately.
- No production or live-attestation claim from drill fixtures: provenance labels and the strict go-live gate (make m42-pilot-go-live) block that by construction.
- No guaranteed-savings claim: the \$1M value architecture is a sponsor-perceived value target; economics are measured per workload against M42's own baseline.
- No universal zkML claim: proof mode is documented per workload.

Reviewer Checklist

For an M42 reviewer who wants to verify this dossier in one sitting:

1. Open any evidence-bundle-<job_id>.json under config/pilots/m42/evidence/exports - confirm hybrid evidence fields, policy block, validator fields, seal_id.
2. Open negative-control-results.json - confirm all six rejection behaviors, including wrong model hash and PHI injection.
3. Open config/pilots/m42/registry/measurements.json - confirm the active model hash matches the bundles.
4. Run make m42-sandbox-preflight - confirm the package validates from a clean checkout.
5. Ask us which of the above a general-purpose chain, a privacy chain, a zkML library, a compute market, or a hyperscaler provides as a native, independently settled object. That question is the dossier.